Yuchen Hou

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DoB: June 2003

EDUCATION BACKGROUND

National University of Singapore **Singapore**
Master of Science in Electrical Engineering Aug 2025 – Now

- **Research Experience:** Core member of the **NUS Advanced Control Lab**, focusing on **Embodied AI** and autonomous robotic decision-making.
- **Research Focus:** Perception and planning for aerial robots based on **Vision-Language Navigation (VLN)**, and the application of **End-to-end Reinforcement Learning** in complex 3D environments.

Nanjing University of Aeronautics and Astronautics **Nanjing, CHINA**
Bachelor of Engineering in Electronic and Communication Engineering Sept 2021 – Jul 2025

- **GPA:** 88/100
- **Key Courses (With Scores):** Advanced Mathematics (96), Probability Theory and Mathematical Statistics (97), Random Signal Analysis (97), Microwave Technology and Antenna (97), Fundamentals of Communication (95), Complex Function (99).

PROJECT EXPERIENCES

LangGap: Diagnosing and Closing the Language Gap in Vision-Language-Action Models
First Author *NUS Advanced Control Lab* Oct 2025 – Now

- *Status:* **Accepted to IROS 2026**.
- *Preprint:* arXiv:2603.00592
- Discovered that state-of-the-art VLA models achieve high benchmark scores but **ignore language instructions**, revealing a gap between performance and true understanding.
- Designed systematic **semantic perturbation** evaluation framework to expose language grounding failures.
- Proposed **multi-task same-scene training** approach and constructed augmented dataset for fine-tuning.

Zero-Shot Indoor Object Navigation for Autonomous Drones Using Dual-Policy Reinforcement Learning
Team Member *NUS Advanced Control Lab* Jun 2025 – Now

- *Status:* **Accepted to IROS 2026**.
- *Preprint:* arXiv:2601.15614
- Developed a dual-mode visual navigation system based on **end-to-end Reinforcement Learning**, utilizing **RGB-D sensor** input.
- Engineered the policy to autonomously switch between **'Exploration'** and **'Target-driven Navigation'** modes based on target visibility.
- Trained the agent from scratch to master a **discrete action space**, including complex maneuvers like take-off, navigation, and landing.

Vision-Language Navigation on Autonomous Drone
Team Member *NUS Advanced Control Lab* Jun 2025 – Now

- *Status:* **Nearing completion**.
- Built a robust pipeline to generate various **3D paths** in the **Habitat simulator**.
- Overcame the challenges of the simulator initially designed only for ground robots by designing a robust **3D navigation algorithm** and obstacle detection method.
- Trained a strong and general policy for **drone navigation**.

Frontier Based Autonomous Exploration Vehicle Design with ROS2 and Lidar
Team leader *NUAA* Sep 2024 – Apr 2025

- Led a team of two members to develop an autonomous exploration system using **ROS2** and **LiDAR** technology.
- Implemented **SLAM** algorithms including **Cartographer** and **Navigation2**, achieving real-time mapping and path planning.
- Integrated **YOLOv11** for object detection.
- Successfully deployed the complete system on an **embedded hardware platform**.

WORK EXPERIENCE

Huawei Norbert Wiener Research Center — **Singapore**
Research Intern — Mar 2026 – May 2026

- Developed backend **deep learning** models for **short-range radar** signal processing.
- Conducted research on downstream tasks including **human pose estimation** and **indoor mapping/SLAM** from radar data.

Movel.AI — **Singapore**
Robotics Software Intern — Dec 2025 – Jan 2026

- Validated autonomous **navigation and control systems** for forklift robots in industrial environments.
- Conducted on-site **factory deployment** and testing in real warehouse settings.
- Integrated robotics software with **fleet management systems**.
- **Tech Stack:** **ROS2**, Navigation Stack, Sensor Integration.

SKILLS & AWARDS

Applied Skills

- **Programming & Systems:** **Python** (Proficient), **Linux (Ubuntu)**, Shell scripting.
- **AI Frameworks & Simulation:** **PyTorch** (DRL policies), **Habitat**, **AI2-THOR** (Complex 3D environments).
- **Robotics Development:** **ROS2** (Nodes, Comm), **SLAM** (Cartographer, Nav2), **YOLO** integration.

Research Capabilities

- **Academic & Theory:** **Sim-to-Real Transfer**, Embodied AI Architectures, **RL Algorithms**.
- **Literature & Writing:** Tracking top-tier conferences (**CVPR/IROS/ICRA**), Academic Paper Writing.

Languages

- **English (Proficient):** IELTS Overall 7.0 (Reading: 8.5, Writing: 7.5, Listening: 6.5, Speaking: 6.0).
- **Mandarin:** Native Speaker.

Awards

- Second-class Scholarship (2024) and Third-class Scholarship (2021, 2022, 2023) at Nanjing University of Aeronautics and Astronautics (NUAA).